

Haonan Jian

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EDUCATION

Brandeis University | Waltham, MA, USA

Sep 2023 – Expected Dec 2026

B.S in Computer Science; Minor in Economics | GPA: 3.7/4 (**Dean's List**)

Relevant courses: Data Structures; Introduction to Machine Learning; Deep Learning; Unsupervised Learning and Data Mining; Automation in Software Development; Introduction to Software Engineering; Operating Systems

SKILLS

- **Programming Languages:** Python, Java, R
- **ML/AI:** PyTorch, TensorFlow, Scikit-learn, NLP, LLM Agents, Model Fine-tuning, RAG
- **Tools & Frameworks:** vLLM, Docker, Pandas, NumPy, Git
- **Cloud & Infra:** AWS

RESEARCH & PROJECTS

Research Assistant

Apr 2025 – Present

Project: Doc2Agent, Supervisor: Professor Pengyu Hong

Doc2Agent builds an automated pipeline that converts unstructured REST API documentation into validated, Python-based tools, enabling LLM agents to call real-world APIs more reliably.

Code repository: github.com/coolkillercat/Doc2Agent

- Ran the full experimental pipeline, executing generated tools across **744 real-world API endpoints**, all WebArena environments, and scientific APIs—achieving a **55.1% improvement** over baseline API-calling agents.
- Deployed the complete WebArena backend stack on **AWS** (AMIs + Docker), configuring security groups and endpoints to support large-scale, reproducible benchmarking.
- Designed and integrated a Python execution module that extends the API agent into a CodeAct-style agent capable of interpreting API docs, issuing Python tool calls, and performing conditional, multi-step reasoning.
- Fine-tuned **Gemma-3** models for structured API information extraction, training the model to identify endpoints, parameters, and usage patterns from unstructured documentation.

AI Engineer Intern

Apr 2026 – June 2026

Company: Yinghan Pharma-Tech (Shanghai) Co., Ltd.

- Extended a **LangGraph-based multi-agent platform** for scientific workflows by integrating a new domain specialist across routing, tool execution, validation, and artifact handling.
- Redesigned an LLM-assisted report-generation workflow with **deterministic interfaces and consistency checks** grounding generated narratives in structured analytical results.
- Developed **Python and R** tooling for covariate-aware scientific analysis and automated quality checks supporting downstream model-selection workflows.

Dominion-Style Card Game (Course Project)

Jan 2025 – May 2025

Team Leader / Developer

Code repository: github.com/brandeis-cosi-103a-spring25/group-b

- Led a 4-person team building a Java Dominion-style card game; implemented core game-state and card-effect logic using **modular OOP** and established GitHub Actions CI with unit and integration tests.

PUBLICATIONS & PRESENTATIONS

- Xinyi, N.^{*}, **Haonan, J.**^{*}, Qiuyang, W., Vedanshi Chentan, S., Pengyu, H.
"Doc2Agent: Scalable Generation of Tool-Using Agents from API Documentation," *AAMAS 2027 submitted*. ^{*} *Equal contribution*.
- Xinyi, N., Qiuyang, W., **Haonan, J.**, Vedanshi Chentan, S., Pengyu, H.
"ToolFactory: Automating Tool Generation by Leveraging LLM to Understand REST API Documentations," *Preprint available on arXiv*.

HONORS

Provost's Summer Undergraduate Research Fellowship, Brandeis University (2026);

Jerome A. Schiff Undergraduate Fellowship, Brandeis University (2025)